

Scalability & Future Plan

Date	02 October 2026
Team ID	SWTID-2026-4644
Project Name	FitBuddy
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Monolithic Architecture	Slow deployment cycles and difficult to isolated faults when one service fails	Medium
2	Single Database Instance	Creates a single point of failure and bottlenecks read/write speeds during peak hours	Medium
3	Lack of Automated Caching	Increases database load and slow down response times for frequently requested data	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single server handling all requests	Implemented a Load Balancer and Horizontal Auto-Scaling to add servers dynamically
2	Data Storage	Hard disk storage on a local DB server	Migrate to a Cloud-based Managed Database with Read Replicas and sharding
3	Performance	High latency during traffic spikes	Introduce an in-memory caching layer using Redis or Memcached
4	Security	Basic firewall and HTTP protocol	Upgrade to HTTPS implement IAM roles, and enforce Data Encryption at Rest

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Containerization with Docker	Month 1-2	Simplifies deployment and ensures consistent environment setups
Phase 3	Database Migration & caching	Month 3-4	Reduces query response times by up to 50% and secure data redundancy
Phase 4	Microservices Translation	Month 5-6	Allows independent scaling of services and improves overall system fault tolerance